

Chatbot Project Presentation

IAM (Identity and Access Management) Request Chatbot



THE UNIVERSITY OF
TENNESSEE
KNOXVILLE

GitHub URL: [cs524-chatbot-project](https://github.com/cs524-chatbot-project)

Presented By: Alex Crumley

Introduction

IAM and Project Motivation

Introduction

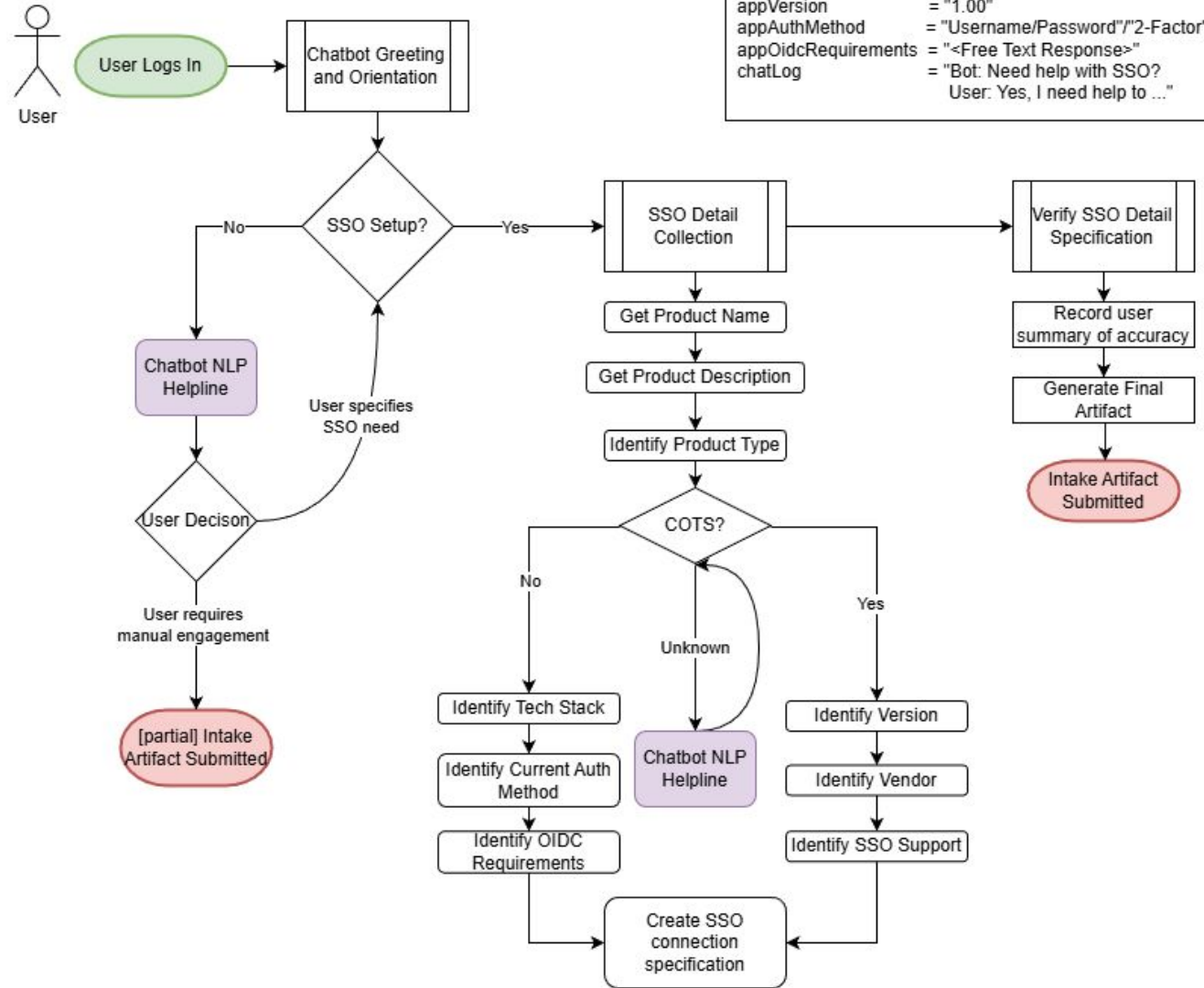
- **IAM (Identity and Access Management)**
 - Managing the entire lifecycle of user identities to ensure the right people have the appropriate access to the right resources.
- **The SSO Setup Problem**
 - Single-Sign-On (SSO) is an industry standard, requiring every modern application to utilize it.
 - IAM teams have to ask the same questions. Over and over.

Chatbot Implementation

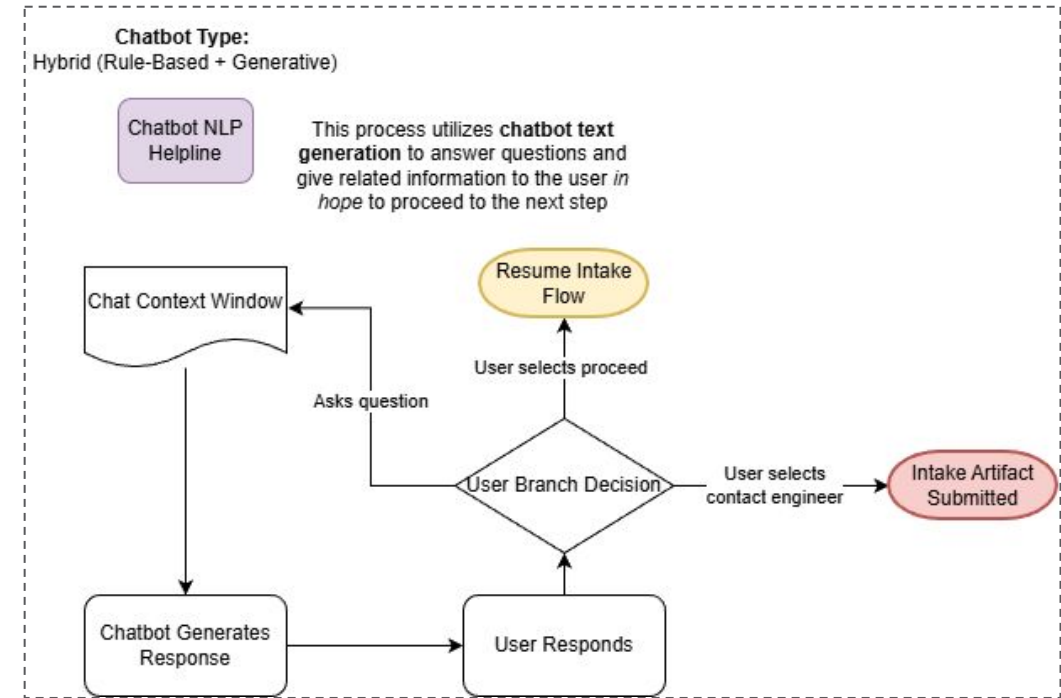
User Interaction, App Architecture

IAM Chatbot Interaction

Defines the ideal interaction of a user with the chatbot's primary feature to engage customer to capture required information for an SSO integration



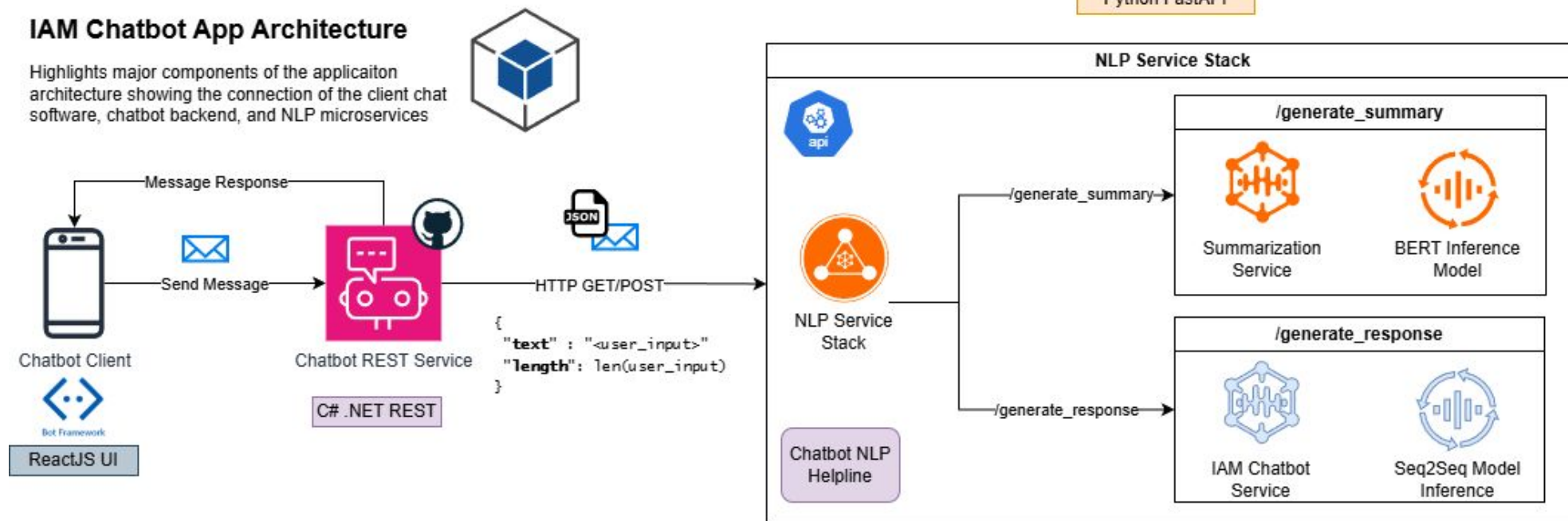
Object: IAM Intake Artifact	
appType	= "COTS"/"CUSTOM"
appName	= "App1"
appDescription	= "<Free Text Response>"
appEnvironment	= "On-Prem"/"Cloud"/"Hybrid"
appVendorInfo	= "<Free Text Response>"
appTechStack	= "React + NextJS + SQL"
appVersion	= "1.00"
appAuthMethod	= "Username/Password"/"2-Factor"
appOidcRequirements	= "<Free Text Response>"
chatLog	= "Bot: Need help with SSO? User: Yes, I need help to ..."



Model Type: Seq2Seq

IAM Chatbot App Architecture

Highlights major components of the application architecture showing the connection of the client chat software, chatbot backend, and NLP microservices



Chatbot Details

Data Collection, Training Pipeline, Evaluation Pipeline

Data Collection

Dataset Overview

- Google search of PDF resources related to IAM
- Performed **NMF** topic modeling to evaluate corpus relevance
- Determined satisfactory relevance

Topic	Interpreted Theme	Top Keywords (NMF)
0	Core IAM Governance	access, identity, management, system, iam, user, security, organization, risk, control
1	Account Provisioning & Mapping	account, mapping, attribute, resource, value, expression, user, synchronization, inbound, source
2	Customer Identity (CIAM)	customer, ciam, experience, solution, modern, wiley, son, ltd, dissemination, john
3	Role-Based Access Control (RBAC)	role, assignment, organizational, policy, privilege, user, assigned, entitlement, structure
4	OT / Network Identity Systems	network, idam, access, pac, ot, cr, component, workstation, rtu, user
5	SSO & Identity Integrations	integration, okta, token, sso, microsoft, app, oin, learn1, authorization, client

IAM Chatbot Training Pipeline

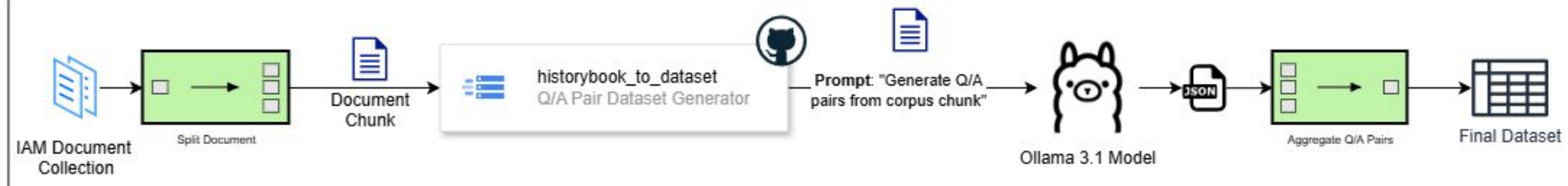
Shows transformation of raw documents (PDFs, markdown files, text files) into Q/A (question/answer) pairs used for training the Seq2Seq model.



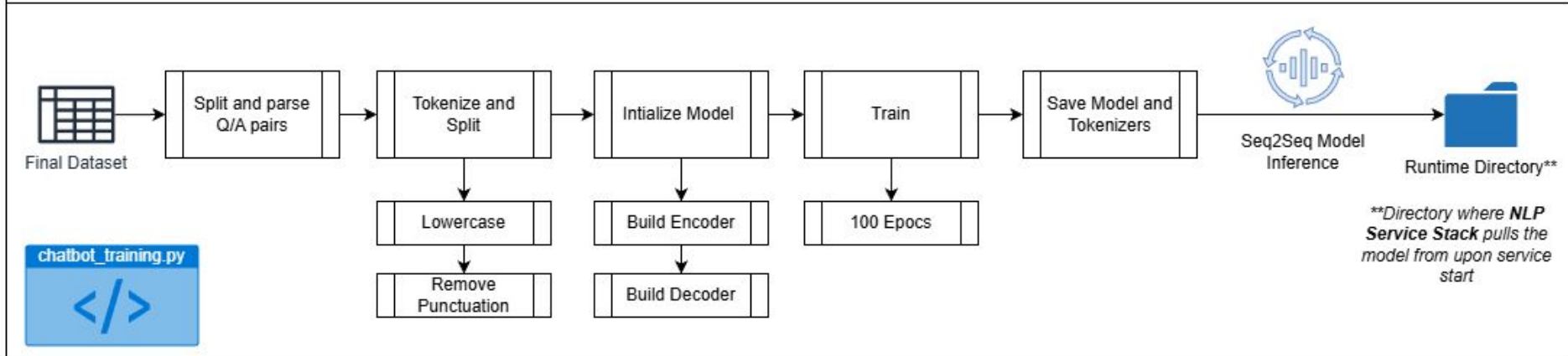
Q/A Pair Example

```
{
  "instruction": "Explain how the identity management system verifies user privilege levels."
  "input": "The identity management system provides data from authoritative sources....."
  "output": "the system provides data from authoritative sources to inform auditors about users."
}
```

Dataset Preparation Pipeline



Chatbot Training Pipeline



IAM Chatbot Evaluation Pipeline

Shows how chatbot performance was evaluated based using the ROGUE metric with the dataset it was trained on.

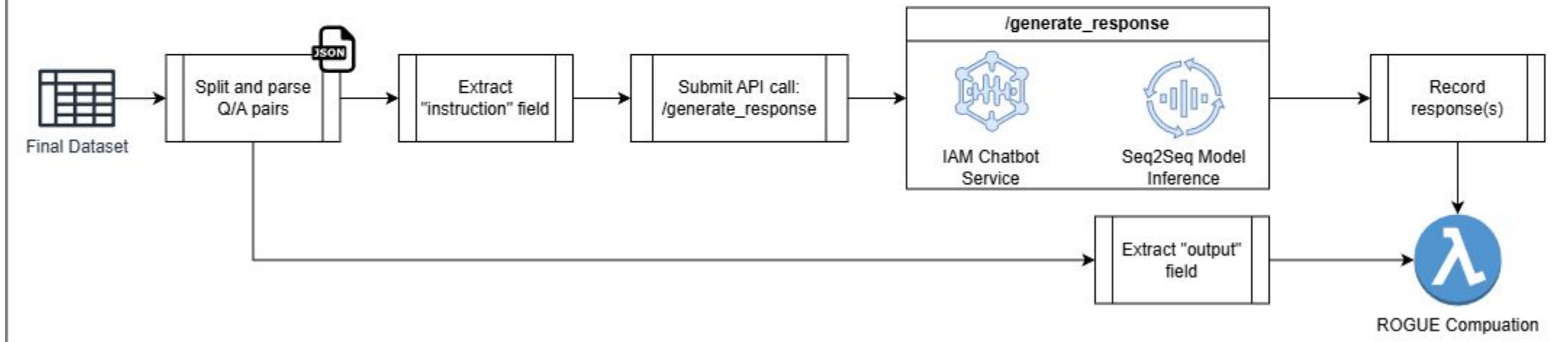


Q/A Pair Example



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Chatbot Evaluation Pipeline



Chatbot Evaluation Results

Evaluation Overview

- Model size and hypertuning parameters greatly influence ROGUE scores
- ROGUE scores prove to be an excellent indicator of chatbot coherency and model overfitting the dataset

Metric	LSTM 256	LSTM 512	GRU 512
ROUGE-1 (F1)	0.3365	0.6530	0.5899
ROUGE-2 (F1)	0.2300	0.5557	0.4910
ROUGE-L (F1)	0.3218	0.6497	0.5834

Demo

User flow interaction (live)

Demo Overview

- **User Interaction**

- Demonstrate the desired user flow to produce an SSO request

- **Chatbot Highlights**

- Demonstrate functionality of rule-based mode for business use-case
- Demonstrate flexibility of generative-based mode for user questions
 - Increases user confidence during interaction and higher-quality artifact output (user inputs the right information and the right time)

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